



TAAM: Inductive Graph-Class Incremental Learning with Task-Aware Adaptive Modulation

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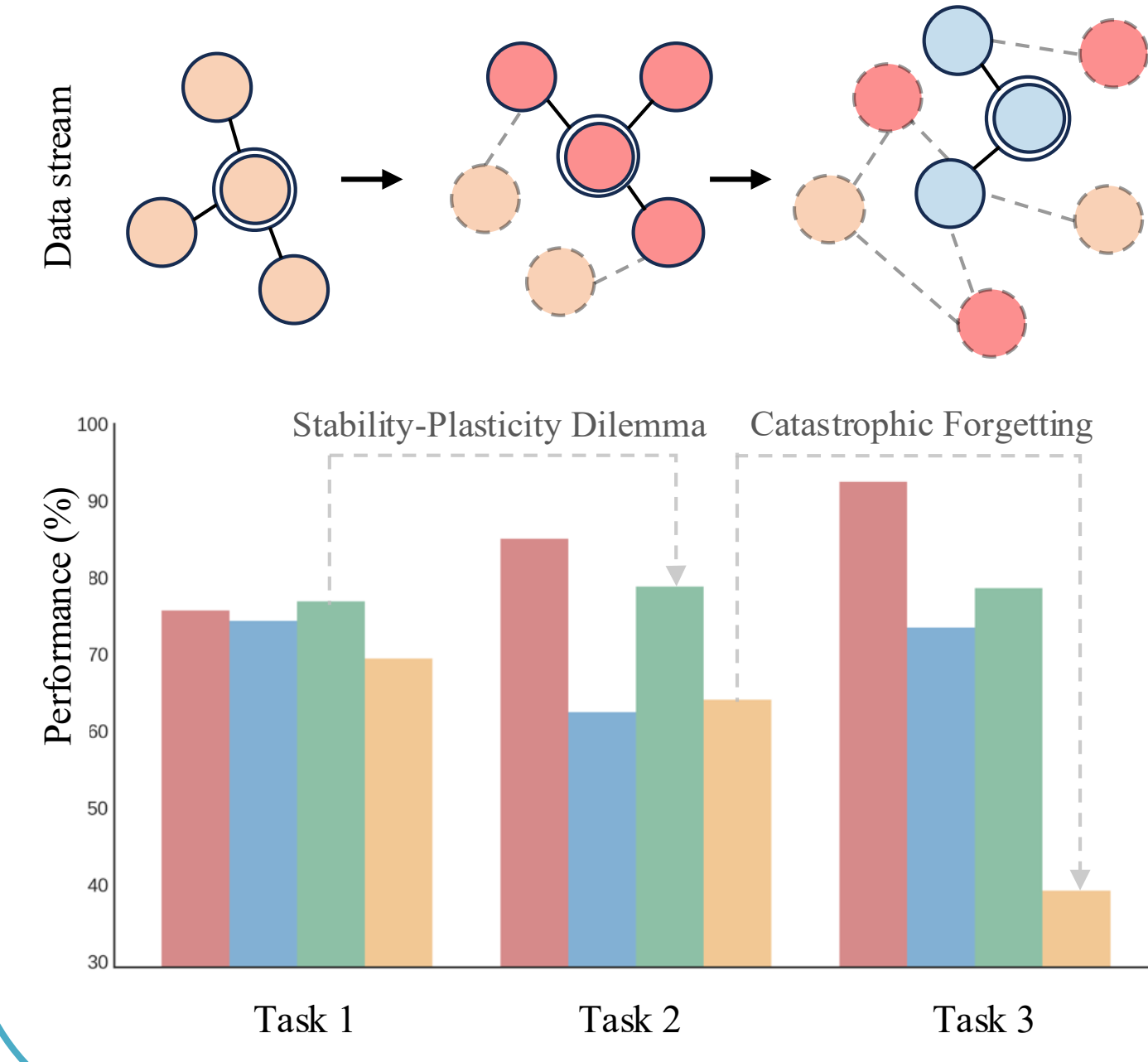


Main Contribution

The main contribution of this work can be summarized as follows:

- We are the first to validate our approach in the more challenging and realistic inductive graph continual learning scenario. **TAAM** operates without reliance on data replay or extensive pre-training, offering a truly resource-efficient solution.
- We propose **Anchored Multi-hop Propagation (AMP)**. It ensures robust task-ID inference, maintains model stability, and prevents catastrophic forgetting across eight datasets.
- We propose a lightweight **Neural Synapse Modulators (NSM)**. Compared to traditional static prompting, it effectively incorporates task-specific knowledge, providing more expressive and fine-grained control over GNN representations.

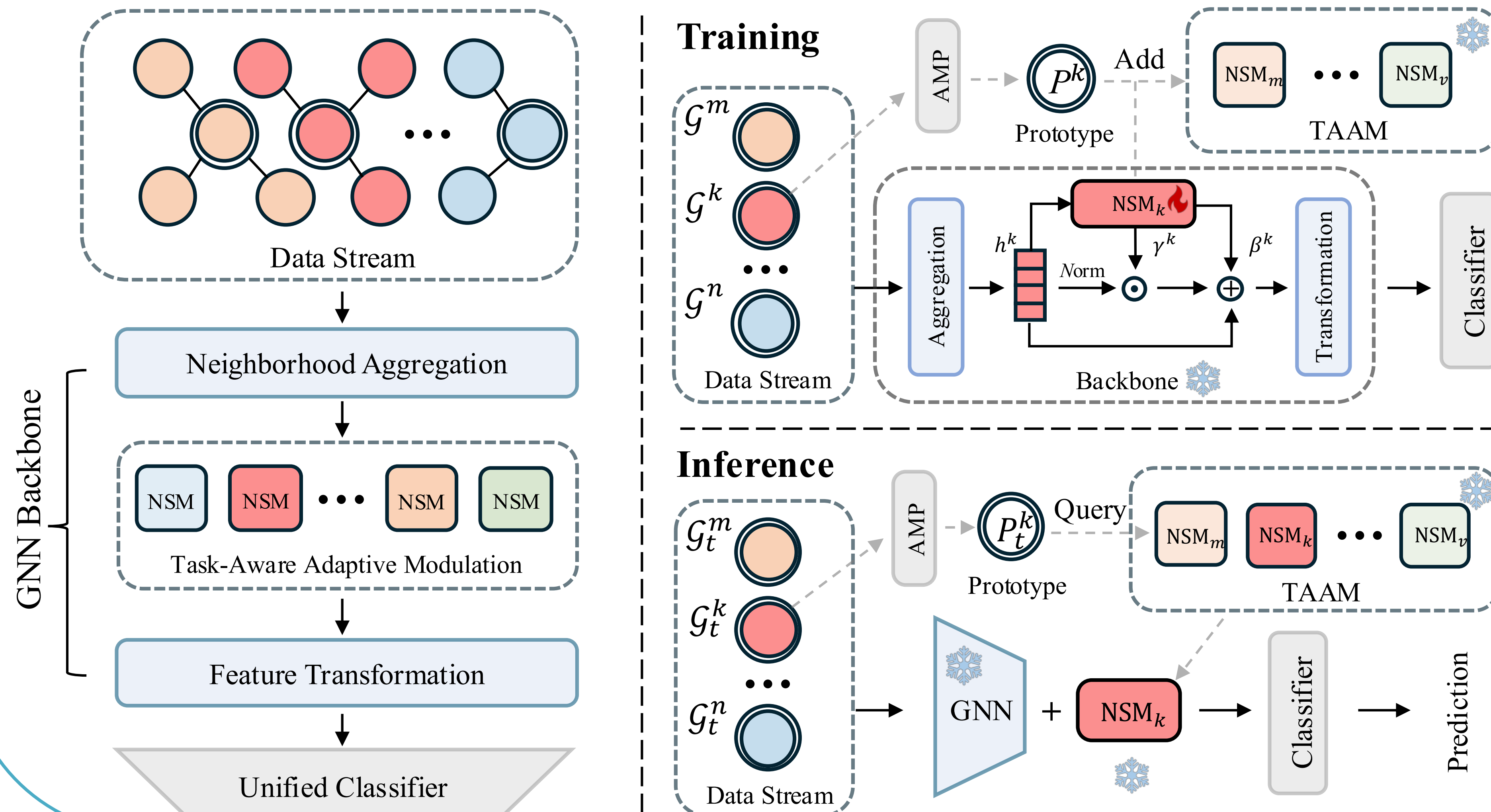
Preliminary



- **Dynamic Graph Streams:** We model continuously evolving real-world graphs as sequential tasks.
- **Stability-Plasticity Dilemma:** Existing baselines fail to balance learning new topologies with retaining prior knowledge.
- **Drawbacks of Replay:** Prevailing methods rely on data replay to mitigate forgetting, incurring severe memory overhead and strict privacy concerns.

Methodology

TAAM, a replay-free and resource-efficient framework for graph continual learning. **TAAM** leverages a frozen GNN backbone and introduces a **lightweight NSM** to dynamically adapt feature representations for each new task. To handle unknown task boundaries during inference, our **AMP** mechanism generates distinct structural prototypes for robust **task-ID inference**, accurately routing target data to the appropriate **expert NSM**.



1 Anchored Multi-hop Propagation (AMP):

$$\mathbf{Z}^{(i+1)} = (1 - \alpha)\mathbf{SZ}^{(i)} + \alpha\mathbf{H}^{(0)}$$

$$\mathbf{H}_{AMP} = \text{CONCAT}(\mathbf{Z}^{(h_1)}, \mathbf{Z}^{(h_2)}, \dots, \mathbf{Z}^{(h_m)})$$

$$\mathbf{P}_k = \frac{1}{|\mathcal{V}_k^{\text{train}}|} \sum_{i \in \mathcal{V}_k^{\text{train}}} (\mathbf{H}_{AMP})_i$$

2 Neural Synapse Modulators (NSM):

$$\mathbf{M}_k = (\mathbf{W}_A \mathbf{e}_k) \cdot \mathbf{W}_B$$

$$\mathbf{a}_i = \text{softmax}(\mathbf{h}_i \mathbf{W}_{att})$$

$$\mathbf{h}'_i = (\gamma_i \odot \text{LayerNorm}(\mathbf{h}_i) + \beta_i) + \mathbf{h}_i$$

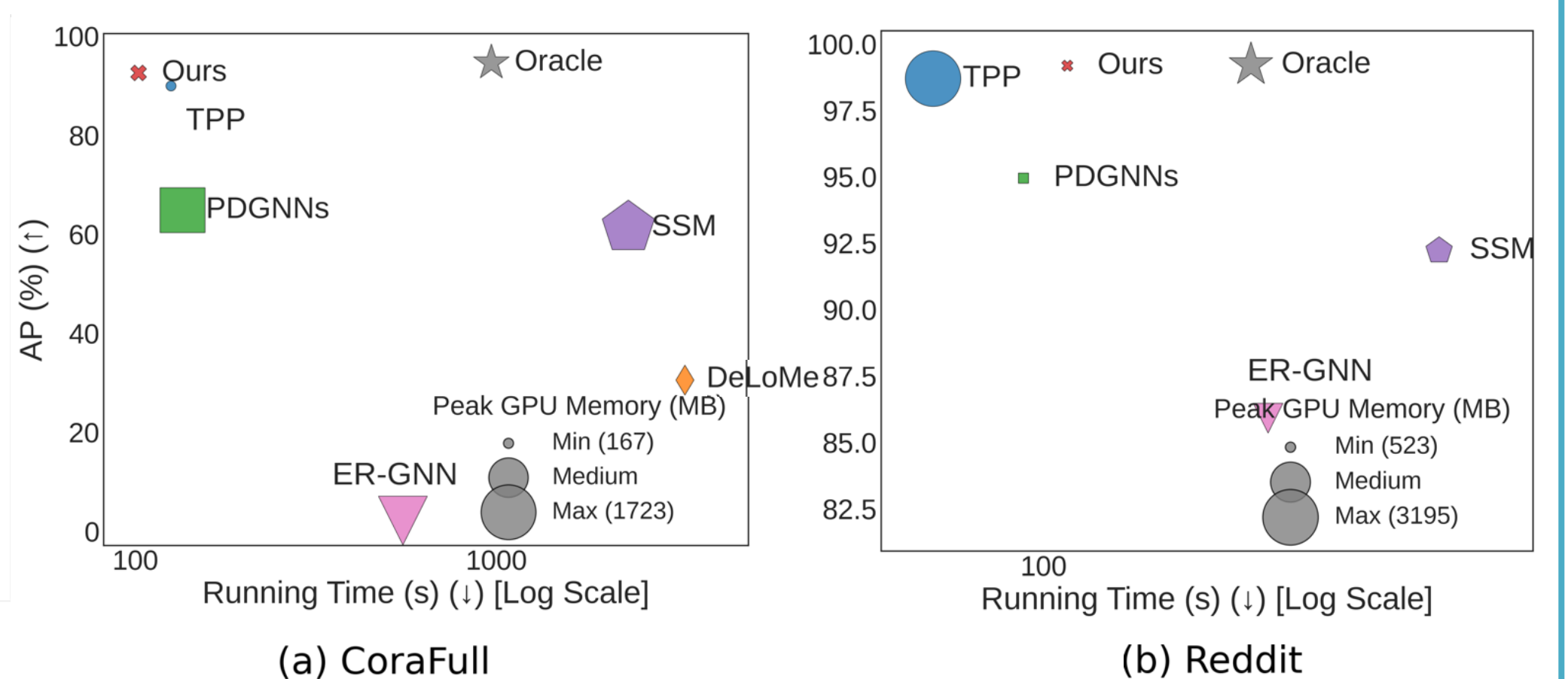
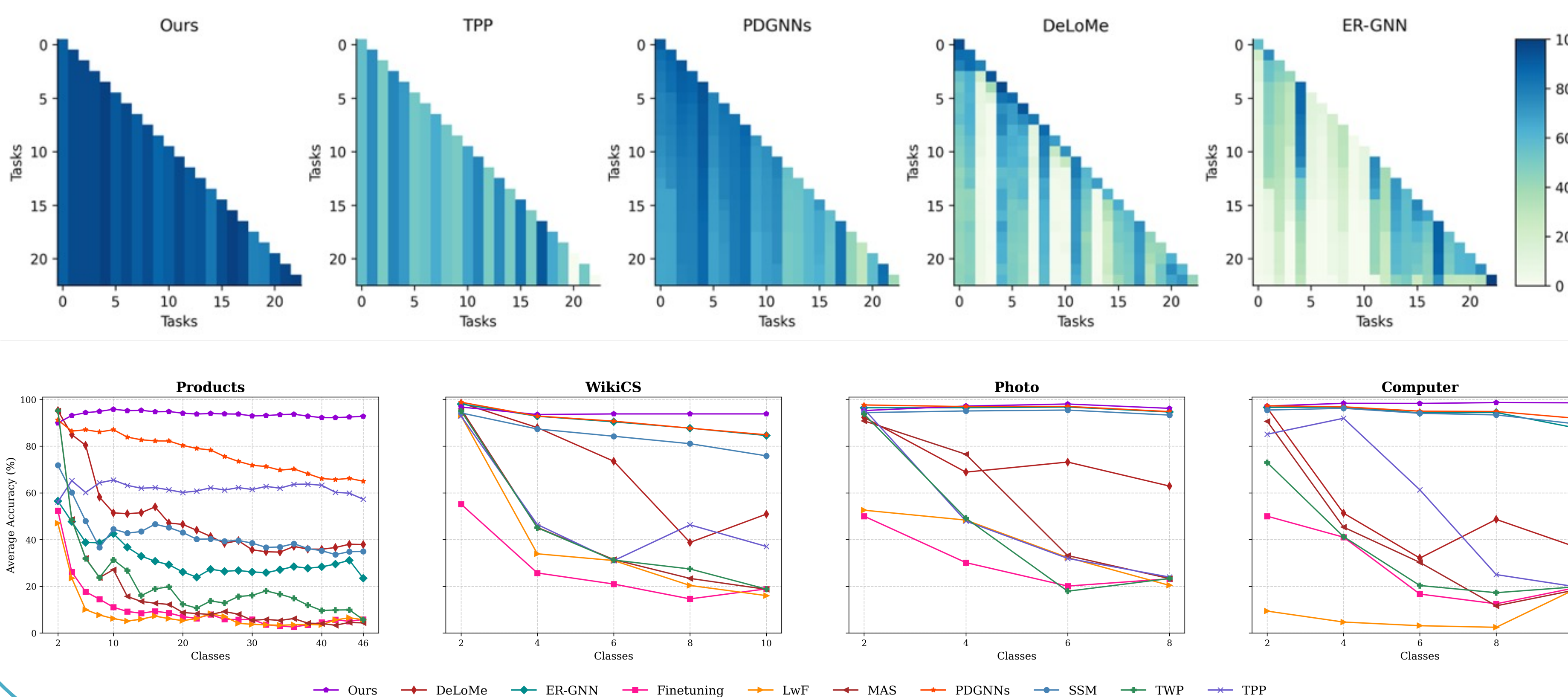
3 Training Objective:

$$\mathcal{L}_{new} = - \sum_{v \in \mathcal{V}_{train}} w_{y_v} \log \left(\frac{\exp(\mathbf{z}_{v,y_v})}{\sum_{c' \in C_{new}} \exp(\mathbf{z}_{v,c'})} \right)$$

Experiment

Category	Model	CoraFull		Arxiv		Reddit		Products		Photo		WikiCS		Computer		Citeseer	
		AA%(↑)	AF%(↓)	AA%(↑)	AF%(↓)	AA%(↑)	AF%(↓)	AA%(↑)	AF%(↓)	AA%(↑)	AF%(↓)	AA%(↑)	AF%(↓)	AA%(↑)	AF%(↓)	AA%(↑)	AF%(↓)
Lower bound	Finetuning	1.8±0.2	13.7±2.5	4.9±0.0	11.8±4.1	4.9±0.0	4.9±0.0	4.7±0.8	31.1±1.6	23.2±0.0	38.6±6.5	18.8±0.1	41.5±7.9	19.0±0.1	28.1±12.0	31.4±0.1	52.3±5.3
	LwF(2017)	1.8±0.2	18.2±2.1	4.9±0.0	30.8±4.8	5.6±0.9	58.4±5.3	21.4±0.9	33.4±0.7	12.5±8.0	46.2±3.1	14.7±1.2	78.4±6.0	8.5±7.8	41.8±32.0	31.4±0.1	70.7±0.8
Regularisation	MAS(2018)	2.8±0.3	67.4±1.4	4.8±0.0	80.9±4.2	5.0±0.1	97.5±0.1	3.9±0.3	88.7±0.4	18.6±0.1	92.3±1.8	23.1±0.2	87.3±14.2	29.9±0.1	80.9±0.4	80.9±0.4	80.9±0.4
	TWP(2021)	7.2±0.9	65.7±1.3	4.9±0.0	60.5±5.3	5.0±0.3	92.8±1.5	5.5±0.2	78.3±1.7	23.2±0.1	73.8±7.5	18.6±0.1	89.2±2.5	19.2±0.3	64.7±15.4	36.5±0.4	66.9±1.2
Replay	ER-GNN(2021)	2.1±0.0	14.1±1.8	16.3±2.7	40.4±2.4	85.0±0.8	12.3±0.8	22.5±0.9	33.1±1.2	94.3±0.2	2.4±0.2	84.1±0.3	7.3±0.5	87.6±0.6	10.6±0.9	61.3±0.4	34.5±0.6
	SSM(2022)	60.3±0.6	11.2±0.3	7.7±2.1	8.8±2.5	91.9±0.3	1.5±0.2	32.9±1.5	15.3±1.6	92.6±0.7	1.9±0.5	74.3±1.6	4.4±1.1	88.7±0.9	3.1±0.8	47.1±4.5	6.5±2.2
Parameter isolation	PDGNNs(2024)	49.8±27.0	11.5±7.9	44.5±0.2	14.2±0.1	94.9±0.1	1.4±0.0	45.2±0.1	7.1±0.4	94.6±0.3	2.2±0.2	84.2±0.5	3.8±0.4	91.4±0.8	5.5±0.9	71.0±0.1	10.9±0.1
	DeLoMe(2024)	30.5±0.1	47.0±0.5	31.0±0.2	24.1±0.2	94.1±0.6	0.2±0.1	37.8±0.5	35.8±0.2	47.4±0.2	64.9±0.3	35.6±0.3	67.1±0.3	37.1±0.1	73.7±0.2	48.4±0.2	32.3±0.2
Upper bound	TPP(2024)	88.9±0.8	0.0±0.0	84.4±1.0	0.0±0.0	98.5±0.2	0.0±0.0	84.9±0.2	0.0±0.0	23.3±0.7	0.0±0.0	36.6±0.5	0.0±0.0	19.9±0.1	46.2±0.5	77.7±1.6	0.0±0.0
	TAAM (Ours)	92.5±0.0	0.0±0.0	88.6±0.4	0.0±0.0	99.0±0.0	0.0±0.0	92.9±0.5	0.0±0.0	96.2±0.4	0.0±0.0	92.8±0.3	0.0±0.0	97.5±0.6	0.0±0.0	83.5±1.0	0.0±0.0
Joint	Oracle	64.1±0.3	-	45.2±0.1	-	95.6±0.1	-	65.2±0.3	-	95.1±0.5	-	83.7±0.3	-	94.9±0.1	-	70.7±0.1	-
	Oracle	94.3±0.3	-	88.5±0.3	-	99.2±0.0	-	94.0±0.6	-	97.0±0.1	-	92.9±0.2	-	97.7±0.1	-	85.2±0.2	-

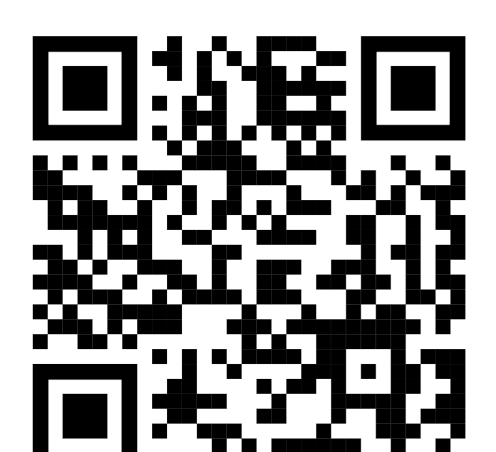
Method	Total Avg AA(↑)	Additional Cost	
		Min.	Max.
Oracle	93.6	2.12 M	82.5638 M
ER-GNN	56.7	1.80 M	103.2285 M
SSM	61.9	0.30 M	51.6099 M
DeLoMe	45.2	0.18 M	34.4037 M
PDGNNs	71.9	1.06 M	100.2838 M
TPP	64.3	0.01 M	1.7949 M
TAAM-L	93.2	0.05 M	9.1812 M
TAAM	92.9	0.02 M	3.0861 M



Paper:



Code:



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